

Crystal structure of a potassium ion transporter, TrkH

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The TrkH/TrkG/KtrB proteins mediate K⁺ uptake in bacteria and probably evolved from simple K⁺ channels by multiple gene duplications or fusions. Here we present the crystal structure of a TrkH from *Vibrio parahaemolyticus*. TrkH is a homodimer, and each protomer contains an ion permeation pathway. A selectivity filter, similar in architecture to those of K⁺ channels but significantly shorter, is lined by backbone and side-chain oxygen atoms. Functional studies showed that TrkH is selective for permeation of K⁺ and Rb⁺ over smaller ions such as Na⁺ or Li⁺. Immediately intracellular to the selectivity filter are an intramembrane loop and an arginine residue, both highly conserved, which constrict the permeation pathway. Substituting the arginine with an alanine significantly increases the rate of K⁺ flux. These results reveal the molecular basis of K⁺ selectivity and suggest a novel gating mechanism for this large and important family of membrane transport proteins.

K⁺ is highly concentrated in all living cells and plays diverse physiological roles such as setting membrane potential¹, regulating turgor pressure² and maintaining intracellular pH^{3–5}. Because K⁺ is virtually impermeable to the cell membrane, specialized K⁺ transporters have evolved to mediate its uptake. In animal cells, K⁺ uptake is mainly achieved by the Na⁺/K⁺-ATPase⁶, whereas in non-animal cells, the task is shared by at least two different systems, one of which, the superfamily of K⁺ transporters (SKT proteins)⁷, is the focus of this research.

An SKT protein has four tandem domains with low homology to each other, each resembling a single protomer from a simple K⁺-channel with a predicted M1–P–M2 transmembrane topology^{8–10}. M1 and M2 are transmembrane helices that are connected by P, the re-entrant pore loop, which is composed of a half-membrane-spanning helix followed by an extended loop¹¹. In K⁺ channels, the half-membrane-spanning helix is called the pore helix, and the extended loop harbours the highly conserved signature sequence TVGYG, which forms the selectivity filter responsible for the coordination of K⁺. In SKT proteins, the equivalent sequences in the P-loops are less strictly conserved. It has been proposed that SKT proteins have a structure that resembles K⁺ channels^{9,10,12}. Results from functional studies have been consistent with this view: mutating a conserved glycine residue in the P-loops changes ion selectivity of SKT proteins in bacteria^{7,13} and plants¹⁴; whereas epitope tagging of a yeast SKT protein showed that its transmembrane topology is consistent with four M1–P–M2 repeats¹⁵.

Studies have shown that the selectivity of K⁺ channels is highly sensitive to changes to the structure of the selectivity filter. Introducing point mutations to the signature sequence of K⁺ channels compromises K⁺

selectivity¹⁶; and the NaK channel, which has a slightly modified signature sequence of TVGDG resulting in an altered selectivity filter conformation, is essentially non-selective between K⁺ and Na⁺ (ref. 17). How, then, can the selectivity of the SKT proteins be maintained, given that their signature sequences are so highly degraded (Supplementary Fig. 1)? Furthermore, if SKT proteins have the architecture of a K⁺ channel pore, how is transport of K⁺ controlled? To address these questions we targeted the bacterial TrkG/TrkH/KtrB proteins, the largest sub-family of SKT proteins, for structural and functional studies. The importance of these proteins in bacteria has been shown in *Escherichia coli*, which does not grow in less than 5 mM K⁺ when its *trkG* and *trkH* genes are deleted¹⁸, and in *Francisella tularensis*, which loses its ability to cause tularaemia when lacking TrkH¹⁹. We crystallized and solved the structure of a TrkH from *V. parahaemolyticus* (hereafter VpTrkH).

Function of VpTrkH

VpTrkH was expressed and purified to homogeneity. The purified VpTrkH eluted as a single symmetrical peak on a size-exclusion column (Supplementary Fig. 2a), and by combining the size-exclusion chromatography with light scattering and refractive index measurements, the molecular mass of the purified protein was estimated to be approximately 100 kDa. Because the molecular mass of each VpTrkH protomer is approximately 54 kDa, the detergent-solubilized VpTrkH is a homodimer. The homodimeric quaternary assembly was further verified by running the protein on an SDS–polyacrylamide gel electrophoresis (SDS–PAGE) gel after incubation with covalent cross-linkers, which produced a band roughly twice the molecular mass (Supplementary Fig. 2b). Dimeric assembly has also been reported

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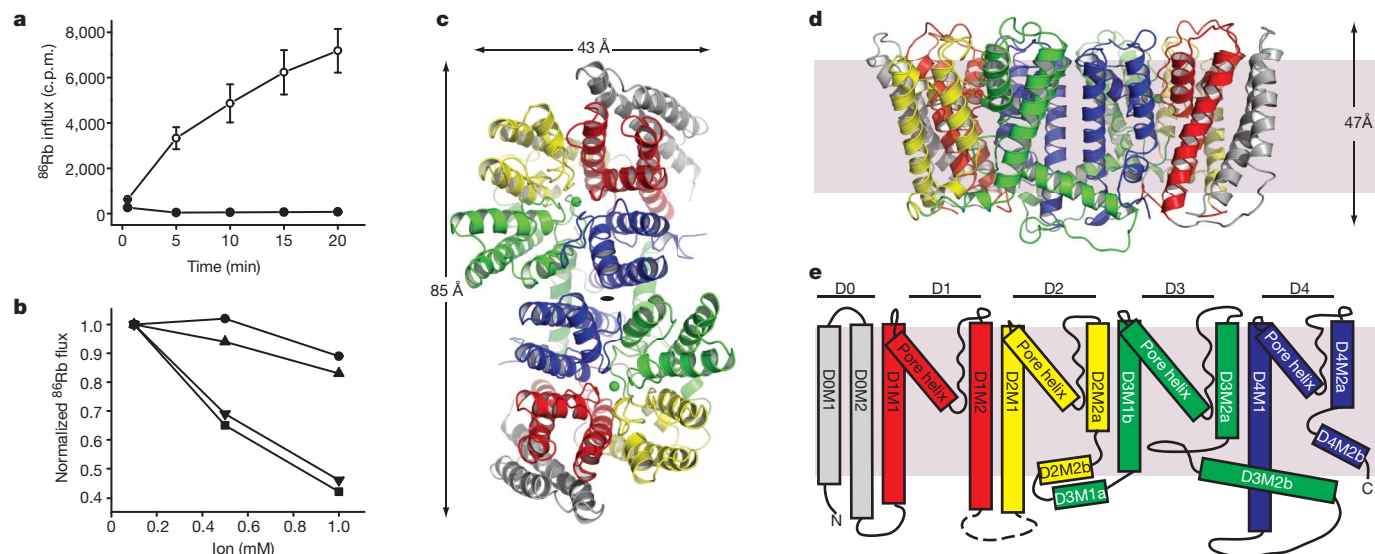


Figure 1 | Function and structure of VpTrkH. **a**, Time-dependent ^{86}Rb influx into liposomes reconstituted with VpTrkH (open circle) or empty vesicles (filled circle). **b**, ^{86}Rb influx at 20 min in the presence of three concentrations of K^+ (square), Rb^+ (inverted triangle), Na^+ (circle) and Li^+ (triangle). **c**, The VpTrkH dimer coloured by domain and viewed from the extracellular side. The twofold symmetry axis is marked as a black oval. The green spheres are K^+ atoms. **d**, VpTrkH viewed from within the membrane

with the extracellular side on top. The dimer is rotated by 90° about the x and y axes relative to **c**. Grey rectangle representing the membrane is shown with a thickness of 30 Å. **e**, VpTrkH topology shown with the extracellular side on top. The five domains are coloured according to the same scheme as in the previous panels. The grey rectangle indicates the thickness of the cell membrane, and the unresolved loop is shown as a dashed line.

for the KtrB protein from *Bacillus subtilis*^{20,21}, a protein closely related to VpTrkH.

Because VpTrkH had not been functionally characterized, we proceeded to measure its function in two experiments. First, the *VpTrkH* gene was introduced into an *E. coli* strain, LB650 (ref. 18), that lacks both the *trkG* and *trkH* genes. LB650 cells have a slow growth phenotype in media with less than 5 mM K^+ , and expression of VpTrkH rescued the growth of these cells (Supplementary Fig. 2c), suggesting that VpTrkH mediates K^+ uptake. Second, purified and detergent-solubilized VpTrkH protein was reconstituted into liposomes, and K^+ permeation was measured indirectly by monitoring uptake of radioactive $^{86}\text{Rb}^+$ (refs 17, 22). In this experiment, the proteoliposomes contain a high internal concentration of K^+ and are diluted into an external solution with a low concentration of K^+ and a trace amount of radioactive $^{86}\text{Rb}^+$. Efflux of K^+ down its chemical gradient via VpTrkH creates an electrical potential across the bilayer that drives uptake of $^{86}\text{Rb}^+$, an ion that is known to permeate K^+ channels. Uptake of $^{86}\text{Rb}^+$ was observed only in vesicles reconstituted with VpTrkH (Fig. 1a), indicating that VpTrkH is permeable to both K^+ and Rb^+ . Further experiments showed that $^{86}\text{Rb}^+$ uptake is inhibited by external K^+ or Rb^+ but relatively unaffected by the presence of Na^+ or Li^+ in the external solution (Fig. 1b). These results indicate that, despite the weak conservation of its signature sequence, VpTrkH exhibits selectivity for K^+ and Rb^+ over Na^+ and Li^+ . In addition, they also suggest that VpTrkH is capable of mediating facilitated diffusion of K^+ driven by an electrochemical gradient.

Overall structure

Crystals of both native and selenomethionine-substituted VpTrkH were grown under oil by the microbatch method, and the best native protein crystals diffracted to 3.5 Å and belonged to the space group $P2_12_1$ (Supplementary Table 1). Initial phases were obtained by single-wavelength anomalous diffraction²³ using a selenomethionine substituted VpTrkH crystal. The building and refinement of the structural model were facilitated by the presence of a twofold non-crystallographic symmetry (NCS) axis and the positions of the selenium atoms. The final refined model contains residues 1–157, 174–484 and one K^+ per subunit. The region encompassing residues

158–173, which is a loop between two transmembrane helices, is disordered.

Each asymmetric unit contains a dimer of VpTrkH protomers related by a twofold symmetry axis perpendicular to the plane of the membrane. Their amino (N) and carboxy (C) termini both probably reside on the cytoplasmic side as inferred from the experimentally determined topology of a highly homologous TrkH protein from *E. coli*²⁴. Two views of the dimer are shown in Fig. 1c, d. Viewed along the twofold axis from the extracellular side, the dimer has a parallelogram shape with sides of approximately 85 and 43 Å. Along the twofold axis, VpTrkH is approximately 47 Å thick. Stereo views of the VpTrkH dimer in three orientations are shown in Supplementary Fig. 3a–c.

Each VpTrkH protomer is composed of five domains, defined sequentially from the N terminus to the C terminus as D0 to D4 (Fig. 1e and Supplementary Figs 1 and 3b, d). D0, which is found only in the TrkH/TrkG subfamily of SKT proteins, has two transmembrane segments. D1 to D4 each have a K^+ -channel-like M1–P–M2 topology, although the M2 helices of D2 to D4 are composed of two shorter ones (Fig. 2a). The secondary structure of each P-loop also resembles that of a K^+ channel, composed of a half membrane-spanning helix, the pore helix, followed by an extended loop that forms the selectivity filter. D1 to D4 assemble around a pseudo-fourfold symmetry axis to form an ion permeation pathway; when observed from the extracellular side along the pseudo-fourfold axis, they are arranged in an anticlockwise direction (Supplementary Fig. 3b). An extensive dimer interface is composed of helices from D3 and D4, with a buried surface of 2,225 Å² per protomer. There is a hydrophobic cavity in the middle of the interface, but it is sealed off from the aqueous medium by two layers of hydrophobic residues on both sides of the membrane (Supplementary Fig. 4). The ion permeation pathway, which is contained within each protomer, has an hourglass shape and two salient features: a selectivity filter and an intramembrane loop (Fig. 2b).

Selectivity filter

The selectivity filter is surrounded by the four pore helices, which are arranged with the negative ends of their helix-dipole moments pointing to the middle of the membrane (Supplementary Fig. 5). A similar arrangement of pore helices is present in K^+ channels, and its role in

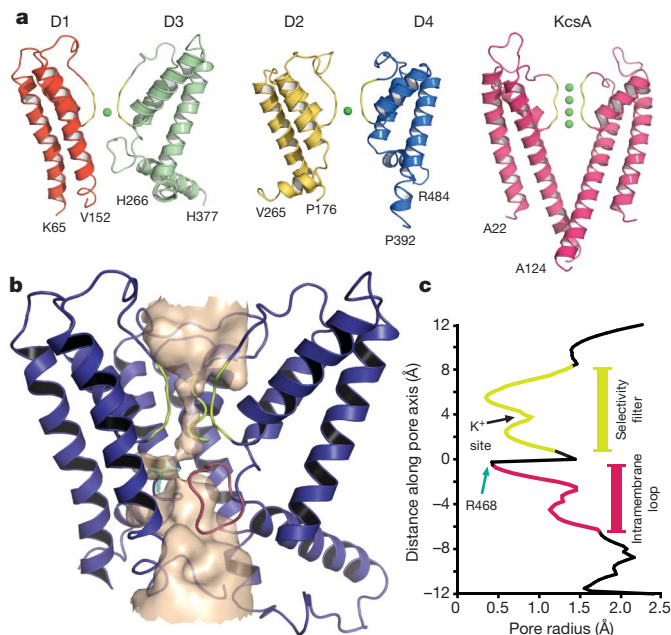


Figure 2 | The VpTrkH pore. **a**, Views of a VpTrkH protomer showing only the D1 and D3 domains (left) or the D2 and D4 domains (middle). Two domains of KcsA are shown on the right for comparison. K^+ atoms are shown as green spheres, and the N- and C-terminal residues are labelled. **b**, Surface representation of the pore of a TrkH subunit obtained with the program Hollow⁴³ using a 1.4-Å probe radius for the vestibules and a 0.75-Å probe radius for the constricted region. The protein is shown with domain 2 removed for clarity and the selectivity filter (yellow), the intramembrane loop (magenta) and residue R468 (teal) highlighted. **c**, Radius of the pore calculated with the program HOLE.

minimizing the free energy of a permeating cation has been discussed for the KcsA K^+ channel^{11,25}.

The four selectivity filter signature sequences are located on the P-loops (Fig. 3a), and these elements come together to form the selectivity filter (omit map in Fig. 3b and Supplementary Fig. 6a, $2F_o - F_c$ map in Supplementary Fig. 6b). Compared with those of K^+ channels, the selectivity filter has a wider opening on the extracellular side and a

much shorter constricted region where K^+ is coordinated (Fig. 3c, d and Supplementary Figs 6c and 7). In the constricted region, main-chain and side-chain oxygen atoms form stacks of oxygen rings that could coordinate and stabilize dehydrated K^+ , a feature that is preserved from the K^+ channels. In an $F_o - F_c$ map calculated with K^+ omitted, two peaks of positive electron density appear in the filter (Fig. 3d and Supplementary Fig. 7a), which we call the upper and lower site and interpret as potential positions where K^+ binds. Owing to the modest resolution of the diffraction data, we cannot unambiguously determine the contribution of K^+ to these densities, as opposed to partial contribution from water molecules or calcium ions that were included in the crystallization solution. Because Rb^+ and Ba^{2+} are known to occupy K^+ positions in the selectivity filter of the KcsA K^+ channel^{11,26}, we took advantage of this knowledge and grew crystals in the presence of Rb^+ or Ba^{2+} (Supplementary Table 1). Difference electron densities corresponding to Fourier coefficients $F_{Rb} - F_K$ or $F_{Ba} - F_K$ are shown in Fig. 3e. In both cases, a strong positive electron density peak is present in the upper site, consistent with substitution of a K^+ with a more electron dense Rb^+ or Ba^{2+} . The upper site lines up with site 3 (S3) in the KcsA K^+ channel, and is constructed entirely by backbone carbonyl oxygen atoms (Fig. 3c and Supplementary Fig. 6c). In the KcsA K^+ channels, Rb^+ does not occupy every K^+ binding site in the selectivity filter, and Rb^+ permeates with a much slower rate than K^+ (refs 27, 28). A slower rate of Rb^+ uptake was also observed by TrkH in *E. coli*²⁹. Therefore, we postulate that the lower peak in the K^+ difference map, which aligns with S4 in KcsA, could potentially be a K^+ binding site (Fig. 3d and Supplementary Fig. 7), although there was no density at this location in either the Rb^+ or Ba^{2+} difference maps.

Although only one binding site has been confirmed by heavy atoms and modelled into the structure, the limited resolution of the data probably reduced our ability to observe more. Comparison with the KcsA structure suggests that the selectivity filter could potentially accommodate a maximum of three K^+ binding sites without requiring a substantial structural change. In addition to the confirmed site and the possible site corresponding to S4 in KcsA, the backbone carbonyls from the highly conserved glycine residues form a ring of oxygen atoms above the confirmed ion-binding site, and could potentially form another K^+ binding site that would line up with S2 in KcsA (Fig. 3d and Supplementary Fig. 7). The S1 site is lost because of the widening of the selectivity filter in TrkH. This is not wholly unexpected owing to

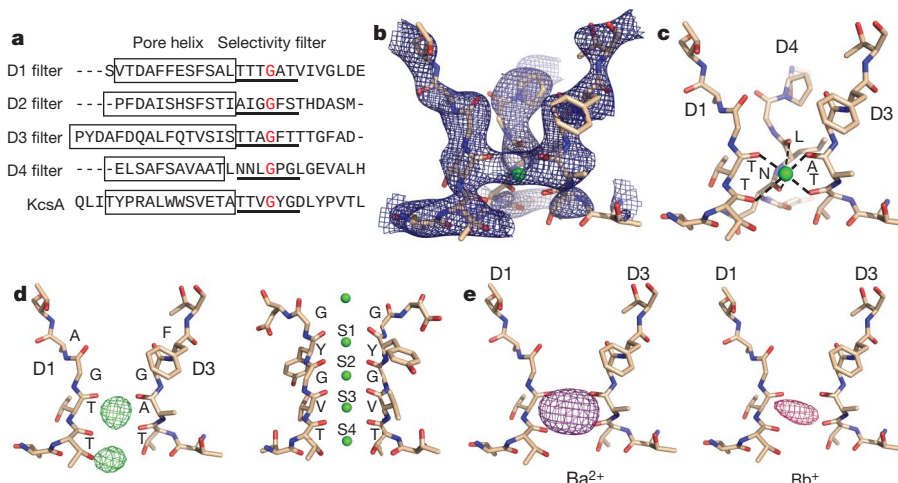


Figure 3 | Selectivity filter of VpTrkH. **a**, Amino-acid sequence alignment of the selectivity filter regions (underlined) and pore helices (box) in VpTrkH with the selectivity filter of the KcsA K^+ channel. The highly conserved glycine residue is marked in red. **b**, **c**, The selectivity filter with domain 2 removed, shown with **b** an NCS-averaged, simulated annealing omit map calculated with six residues from each selectivity filter omitted, contoured at 1σ , or **c** the coordination geometry of the K^+ (green sphere) highlighted. **d**, D1 and D3

from the K^+ structure are shown with $F_o - F_c$ electron density calculated without K^+ in the model and contoured at 3.5σ . The filter of KcsA is shown on the right for comparison. **e**, Ion-binding sites in the selectivity filter. Ba^{2+} (left) and Rb^+ (right) ($F_{o(ion)} - F_{o(K)}$) difference Fourier maps are shown contoured at 6.0 and 3.5σ levels, respectively, calculated using phases from the K^+ structure. The stick models are D1 and D3 from the K^+ structure.

the differences in the selectivity filter sequence between TrkH and KcsA: the second glycine residue in the signature sequence of K⁺ channels is not conserved in TrkH (Fig. 3a), and in the high-resolution KcsA structure³⁰ the backbone torsion angles of this residue lie in an unfavourable region of the Ramachandran plot for non-glycine residues. Regardless of the number of sites, the structure shows that at least one dehydrated K⁺ binds to VpTrkH through coordination by oxygen atoms in a manner similar to a K⁺ channel. Further experiments are needed to measure more accurately the occupancy of K⁺ and whether VpTrkH exhibits the high selectivity characteristic of K⁺ channels. In addition, because the four-site configuration is crucial for a high K⁺ throughput in K⁺ channels^{27,31}, we conjecture that VpTrkH, if it operates by a channel-like mechanism, conducts K⁺ with a lower throughput.

The ion permeation pathway

Although each protomer forms a continuous pore, a K⁺ permeating from the extracellular side encounters a barrier shortly after it exits the selectivity filter (Fig. 2c). The barrier is composed of two elements: Arg 468 from the second transmembrane segment of domain 4 (D4M2), and an intramembrane circular loop between D3M2a and D3M2b (Figs 2b and 4a). This is a unique feature and is not observed in any known K⁺ channel structures.

Arg 468 is conserved in almost all bacterial SKT proteins. The guanidinium group points towards the centre of the permeation pathway and is approximately 3.1 Å away from the backbone carbonyl oxygen atom of Gly 353, which is part of the intramembrane loop formed by Gly 346 to Lys 357 (Fig. 4a). This loop is present in all TrkH, TrkG and KtrB families, is rich in glycine and other small residues such as alanine and serine, and contains several highly conserved residues (Supplementary Figs 1 and 8). In addition, Glu 470,

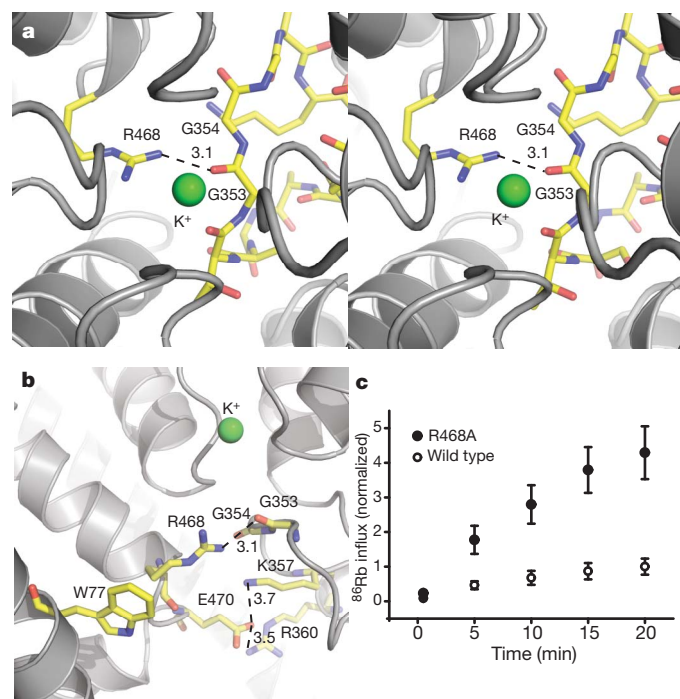


Figure 4 | Constriction formed by Arg 468 and the intramembrane loop. **a**, Stereo view of the interactions between the intramembrane loop and Arg 468 as viewed looking down the selectivity filter from the extracellular side. **b**, Interactions between the intramembrane loop and Arg 468 and Glu 470 as viewed from within the plane of the membrane. Residues Gly 353–354, Lys 357, Arg 360, Arg 468 and Glu 470 are shown as stick representations, and the dashed lines indicate distances between them. **c**, Time-dependent ⁸⁶Rb influx into proteoliposomes reconstituted with wild-type (open circle) or R468A (filled circle) VpTrkH. Error bars, s.e.m.

another highly conserved residue on D4M2, is close to Lys 357 in the loop and Arg 360 in D3M2b: a carboxylate oxygen on Glu 470 is 3.7 and 3.5 Å away from the terminal nitrogens of Lys 357 and Arg 360, respectively. These electrostatic interactions probably further stabilize the position of the intramembrane loop (Fig. 4b and Supplementary Fig. 9).

The narrow constriction formed by Arg 468 and the intramembrane loop has to widen for a K⁺ to reach the intracellular side. To understand further the role of Arg 468 in K⁺ permeation, we substituted it with an alanine and reconstituted the R468A mutant protein into liposomes. ⁸⁶Rb⁺ flux was then measured both for the wild type and for the R468A mutant in side-by-side experiments. Tracer uptake by the R468A mutant is significantly faster than that by the wild type (Fig. 4c), consistent with the observation that R468 occludes K⁺ permeation. An earlier study showed that mutating the equivalent of Arg 468 in a KtrB reduces K⁺ uptake in *E. coli*³², which seems to contradict our functional results and the role of Arg 468 inferred from the structure. However, because the KtrB mutant was assayed *in vivo*, additional factors such as expression levels or association with auxiliary proteins could have affected the measurement. As for the role of the intramembrane loop, a recent study showed that various deletions of the corresponding loop in a KtrB significantly increase K⁺ transport activity³³, consistent with its position shown by the structure.

Discussion

The pseudo-symmetry arising from duplication of an ancestral channel that is observed in the SKT family is not unique among ion channel/transporter families. In the animal kingdom, gene fusion or duplication of a more complex potassium channel, the voltage-dependent K⁺ channel (K_v), generated voltage-dependent Na⁺ and Ca²⁺ channels (Na_v and Ca_v, respectively)^{34,35}. Similar to the SKT proteins, four pseudo-subunits of K_v channels are expressed as a single polypeptide chain in the Na_v and Ca_v channels, although the domains are assembled in a clockwise direction when viewed from the extracellular side^{36,37} in contrast to the anticlockwise orientation observed in VpTrkH. The pseudo-fourfold symmetry in VpTrkH is strongest at the ion-binding sites in the selectivity filter region, where each domain contributes similarly to coordination of a K⁺. However, the symmetry starts to break down outside the selectivity filter where the pore helices vary in length and form different angles with the membrane norm, and the symmetry becomes considerably weaker for the M1 and M2 helices from different homologous domains. Especially notable is D3, which contains the intramembrane loop and the tilted D3M2b helix, and therefore is expected to contribute to gating of the permeation pathway more than the other domains. It is probable that in Na_v and Ca_v channels, the fourfold symmetry is maintained at the selectivity region for coordination of permeating ions but gradually breaks down so that the voltage-sensing modules and the connected gating machinery of a particular pseudo-subunit may contribute more than the others.

TrkH/TrkG and KtrB assemble with the cytosolic adenine nucleotide-binding proteins TrkA^{38,39} and KtrA⁴⁰, respectively. KtrA forms a ring and can undergo significant conformational changes upon binding to, or changes in the oxidation state of, ligands such as NAD and ATP^{20,41,42}. The ring has a twofold symmetry that matches that of the homodimeric assembly of the transmembrane subunits, and could potentially allosterically control K⁺ permeation. We speculate that the dimeric assembly is required for regulation of K⁺ transport by TrkA, although each protomer contains an independent ion permeation pathway. Although the structure of VpTrkH alone does not answer the question of whether TrkH operates as a channel or a transporter, it provides a framework for further studies that will reveal the molecular mechanism of K⁺ uptake and its regulation by the intracellular TrkA subunit.

METHODS SUMMARY

Both native and selenomethionine-substituted VpTrkH were expressed in *E. coli* with a C-terminal polyhistidine tag, extracted into decylmaltoside, and purified

by a metal-affinity column. The proteins were further purified by size-exclusion chromatography and concentrated to 8–10 mg ml⁻¹ for crystallization trials. Crystals were obtained by microbatch crystallization under paraffin oil. Co-crystallization with different ions was achieved by running the size-exclusion chromatography in buffers containing 150 mM of KCl or RbCl, or co-crystallization with BaCl₂. Initial phases were obtained by single-wavelength anomalous diffraction from a 3.9-Å data set collected on selenomethionine-derivatized VpTrkH, and a 3.5-Å native data set (with K⁺) was used for refinement of the final model. The final values of *R* and *R*_{free} were 24.9% and 29.9%, respectively. Purified VpTrkH was reconstituted into liposomes for ⁸⁶Rb⁺ flux assays as described previously^{17,22}.

Full Methods and any associated references are available in the online version of the paper at www.nature.com/nature.

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Supplementary Information is linked to the online version of the paper at www.nature.com/nature.

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Author Contributions M.P., J.L., B.R. and W.A.H. identified TrkH/TrkG/KtrB homologues in the database. R.B., B.K. and J.L. cloned and tested expression of the homologues. Y.C., H.H., J.W., E.J.L. and M.Z. scaled up production of proteins, produced and refined VpTrkH crystals, and collected and analysed X-ray diffraction data. X.J., E.J.L. and M.Z. solved and refined the structures. V.K., S.Y. and E.M.-H. analysed diffraction data and obtained a partial model in early stages of the project. Y.C., M.G.D., M.Q., Y.P., Y.J., J.A.J. and M.Z. characterized VpTrkH function. K.R.R. and W.A.H. advised on data collection and crystallography. E.J.L. and M.Z. wrote the manuscript with inputs from all authors.

Author Information Atomic coordinates and structure factors are deposited in Protein Data Bank under accession number 3PJZ. Reprints and permissions information is available at www.nature.com/reprints. The authors declare no competing financial interests. Readers are welcome to comment on the online version of this article at www.nature.com/nature. Correspondence and requests for materials should be addressed to M.Z. (mz2140@columbia.edu).

METHODS

Homology screen, cloning and initial protein expression. TrkH was first established to be a valid target for structural studies by a bioinformatics analysis^{44,45}. A total of 91 *trkG/trkH/ktrB* genes from 58 prokaryotic genomes were identified, and the genes were amplified by PCR from the genomic DNAs, inserted into a modified pET plasmid (Novagen) with a C-terminal deca-histidine tag and a TEV protease recognition site, and expressed in a small-scale culture. Protein expression was then examined using western blots as a readout, and western-positive clones were pursued for further study. Identification and cloning of homologues, and the initial expression study, were performed by a high-throughput approach in the central facility of the New York Consortium on Membrane Protein Structure; a detailed description of the procedures can be found in ref. 44.

Large-scale protein expression, purification and crystallization. Western-positive clones received from the New York Consortium on Membrane Protein Structure were scaled up for mid- to large-scale expression studies. Five proteins (TrkHs from *V. parahaemolyticus*, *Vibrio fischeri*, *Idiomarina loihiensis*, *Campylobacter jejuni* and a KtrB from *V. fischeri*) had yields higher than 0.25 mg l⁻¹ cell culture and three of them (TrkHs from *V. parahaemolyticus*, *I. loihiensis* and *C. jejuni*) exhibited a mono-dispersed profile in size-exclusion chromatography. Among those proteins, only TrkH from *V. parahaemolyticus* (VpTrkH) produced diffracting crystals and thus became the focus of crystallization efforts.

For large-scale purification of native VpTrkH, BL21(DE3) cells were grown in Luria broth at 37 °C and induced with 0.5 mM isopropyl β -D-1-thiogalactopyranoside (IPTG) after absorbance ($A_{600\text{ nm}}$) reached 1.0; for expression of selenomethionine-incorporated proteins, the cells were grown in minimal medium containing 32.2 mM K₂HPO₄, 11.7 mM KH₂PO₄, 6 mM (NH₄)₂SO₄, 0.68 mM Na citrate, 0.17 mM Mg₂SO₄, 32 mM glucose, 0.008% (w/v) alanine, arginine, aspartic acid, asparagine, cysteine, glutamic acid, glycine, histidine, proline, serine, tryptophan, glutamine and tyrosine, 0.02% (w/v) isoleucine, leucine, lysine, phenylalanine, threonine and valine, 25 mg l⁻¹ L-selenium-methionine, 32 mg l⁻¹ thiamine, and 32 mg l⁻¹ thymine, and induced at an absorbance ($A_{600\text{ nm}}$) of 0.6. The cell membranes were solubilized with 40 mM *n*-decyl- β -D-maltoside (Anatrace) and the His-tagged protein was purified with TALON Metal Affinity Resin (Clontech). After removal of the tag with tobacco etch virus protease, the native protein was subjected to size-exclusion chromatography with a Superdex 200 10/300 GL column (GE Health Sciences) equilibrated in a buffer of 150 mM KCl, 20 mM HEPES, pH 7.5, 5 mM β -mercaptoethanol and 3.5 mM *n*-decyl- β -D-maltoside. The selenomethionine-incorporated VpTrkH protein was purified by the same procedure. The native protein was concentrated to 8 mg ml⁻¹ and the selenomethionine-substituted protein to 10 mg ml⁻¹ as approximated by ultraviolet absorbance.

Although both the native and selenomethionine-substituted VpTrkH yielded crystals readily, the crystals diffracted anisotropically and an overwhelming majority failed to reach 4.0 Å. More than 3,000 crystals were screened over a period of over 3 years. Native VpTrkH crystals were grown by microbatch crystallization under paraffin oil where 1.5 μ l of the protein solution was mixed with an equal volume of crystallization solution containing 35% PEG400, 200 mM calcium acetate and 100 mM sodium acetate, pH 5.3. Rb⁺-derivatized crystals were obtained by the same method except that the size-exclusion chromatography during purification was conducted in a buffer using 150 mM RbCl to replace KCl. Ba²⁺-derivatized crystals were obtained by adding 10 mM BaCl₂ into the native protein before mixing with crystallization solution. Before flash-freezing in liquid nitrogen, the crystals were cryoprotected in mother liquor for 2–5 s. The mother liquor was obtained by vapour diffusion in sitting drops mixed from 3 μ l of the protein solution and an equal volume of well solution containing 35% PEG400, 200 mM calcium acetate and 100 mM sodium acetate, pH 5.3.

Data collection and structure solution. Diffraction data were collected on beamlines X25 and X29 at the National Synchrotron Light Source and 24ID-C and 24ID-E at the Advanced Photon Source. A 3.9-Å data set was collected at a wavelength of 0.9791 Å on selenomethionine-derivatized VpTrkH. The data were indexed, integrated and scaled using HKL2000 (ref. 46). The data set showed anisotropy, but nonetheless exhibited a strong anomalous signal and was therefore used to obtain the initial phases to 3.9 Å by single-wavelength anomalous diffraction. The positions of 18 out of 24 available selenium sites in the asymmetric unit were located using the program phenix.hyss⁴⁷. Initial experimental phases were improved by twofold NCS averaging and solvent flattening using RESOLVE⁴⁷. The resultant density-modified experimental map was used to build manually a partial C α trace with COOT⁴⁸. The phases were gradually extended to a higher-resolution native data set at 3.5 Å using twofold NCS averaging, solvent flattening and histogram matching in DM⁴⁹. Manual model building and sequence assignments were done iteratively using COOT, and the structure

refinement was done using PHENIX⁴⁷ with strong twofold NCS restraints. The final refined model contained residues 1–157, 174–484, and one K⁺ in each subunit in the asymmetric unit. The anomalous difference Fourier map calculated with phases from the refined model confirmed the correctness of the initial selenium sites, all of which overlay well with ordered methionine residues in the final model, and identified two additional sites corresponding to two N-terminal methionines in the asymmetric unit (Supplementary Fig. 10a–c). The region encompassing residues 158–173 is disordered, consistent with the absence of anomalous peaks expected from Met 158 and Met 174 within this region. At the model-building stage, diffraction data were corrected for anisotropy using the Anisotropy Server⁵⁰ and a second model was refined with anisotropy correction. Models with or without the correction essentially overlap; however, map quality is improved in several regions after the correction.

The final refined model devoid of K⁺ was used to calculate the $F_{\text{Rb}} - F_{\text{K}}$ and $F_{\text{Ba}} - F_{\text{K}}$ difference maps for structures containing Rb⁺ and Ba²⁺.

E. coli complementation assay. *E. coli* strain LB650 competent cells were transformed with pET31 vector carrying *V. parahaemolyticus trkH*. Two media, Hi-K and Lo-K, were prepared based on ref. 51, both containing 8 mM ammonium sulphate, 0.4 mM magnesium sulphate, 1 mM sodium citrate, 1 mg l⁻¹ thiamine and 2 g l⁻¹ glucose. For the Hi-K medium, 115 mM potassium phosphate (pH 7.0) was added; for the Lo-K medium, 115 mM sodium phosphate (pH 7.0) was added. The two solutions were mixed in different ratios to produce the desired K⁺ concentration. The transformation cell culture was spread onto agar plates prepared with solutions with different K⁺ concentrations and incubated at 37 °C overnight. As a blank control, pET31 vector without insertions was used to transform *E. coli* strain LB650 competent cells.

Reconstitution of TrkH into proteoliposomes. Purified VpTrkH was reconstituted into lipid vesicles composed of 1-palmitoyl-2-oleoyl-phosphatidylethanolamine and 1-palmitoyl-2-oleoyl-phosphatidylglycerol (Avanti Polar Lipids) in a ratio of 3:1 by mass, as previously described⁵¹. The protein to lipid ratio was 1:150 by mass. The solution was then subjected to several rounds of dialysis against reconstitution solution until the detergent was totally removed. At the end of each experiment, valinomycin was added to the reaction mixture to monitor the maximum uptake.

Determination of protein oligomeric state. The mass of the VpTrkH protein in solution was measured using a Wyatt miniDAWN TREOS 3 angle-static light-scattering detector, a Wyatt Optilab rEX refractive index detector and an Agilent Variable Wavelength Detector ultraviolet absorbance detector⁵². Purified protein sample (5 μ l) was injected onto a TSKgel SuperSW3000 (4.6 mm internal diameter $\times$ 30 cm) silica-gel size-exclusion column in buffer containing 0.016% dodecyl-maltoside at a rate of 0.25 ml min⁻¹. The differential refractive index (dn/dc) value for dodecylmaltoside was calculated to be 0.128 ml g⁻¹ using the Wyatt Optilab rEX refractive index detector. Deconvolution of the protein–detergent conjugate was then achieved using a previously described method⁵³. The calculation did not account for refractive index contributions due to bound lipid.

⁸⁶Rb flux assay. The ⁸⁶Rb⁺ flux assay was performed as described previously²². In the competition assays, Li⁺, Na⁺, K⁺ or Rb⁺ were added directly into the flux buffer, and the readings were taken at the 20 min time point.

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